

Aim:-

- 4 Implementation of decision tree classification algorithm using any dataset available in public domain.

Source code

```
import pandas as pd
import numpy as np
from sklearn.datasets import load_iris
data = load_iris()
print(data.data.shape)
print("classes to predict:", data.target_names)
x = data.data
y = data.target
display(x.shape, y.shape)
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
xtrain, xtest, ytrain, ytest = train_test_split(x, y,
random_state=50, test_size=0.25)
classifier = DecisionTreeClassifier()
classifier.fit(xtrain, ytrain)
ypred = classifier.predict(xtest)
```

```
from sklearn.metrics import accuracy_score
print("Accuracy on train data using Gini:")
accuracy_score(y_true=ytrain, y_pred=
classifier.predict(xtrain)))
print("Accuracy on test data using Gini:")
accuracy_score(y_true=ytest, y_pred=ytest)
classifier_entropy = DecisionTreeClassifier(
    criterion='entropy')
classifier_entropy.fit(xtrain, ytrain)
ypred_entropy = classifier_entropy.predict(xtest)
print("Accuracy on train data using entropy:")
accuracy_score(y_true=ytest, y_pred=ytrain,
y_pre ypred_entropy) y_pred = classifier_entropy.
predict(xtrain)))
print("Accuracy of test data using entropy:")
accuracy_score(y_true=ytest, y_pred =
ypred_entropy))
classifier_entropy1 = DecisionTreeClassifier(
    criterion='entropy', min_samples_split=50)
classifier_entropy1.fit(xtrain, ytrain)
```



```
y_pred_entropy1 = classifier_entropy1.predict(x_test)
print("Accuracy on train data using  
entropy:", accuracy_score(y_true=y_train,  
y_pred=classifier_entropy1.predict(x_train)))
```

```
print("Accuracy on test data using  
entropy:", accuracy_score(y_true=y_test, y_pred=  
y_pred_entropy1))
```

```
from sklearn.tree import export_graphviz
```

```
from six import StringIO
```

```
from IPython.display import Image
```

```
import pydotplus
```

```
dotdata = StringIO()
```

```
export_graphviz(classifier, out_file=dotdata,  
filled=True, rounded=True, special_characters=  
True, feature_names=data.feature_names,  
class_names=data.target_names)
```

```
graph = pydotplus.graph_from_dot_data(  
dotdata.getvalue())
```

```
Image(graph.create_png())
```